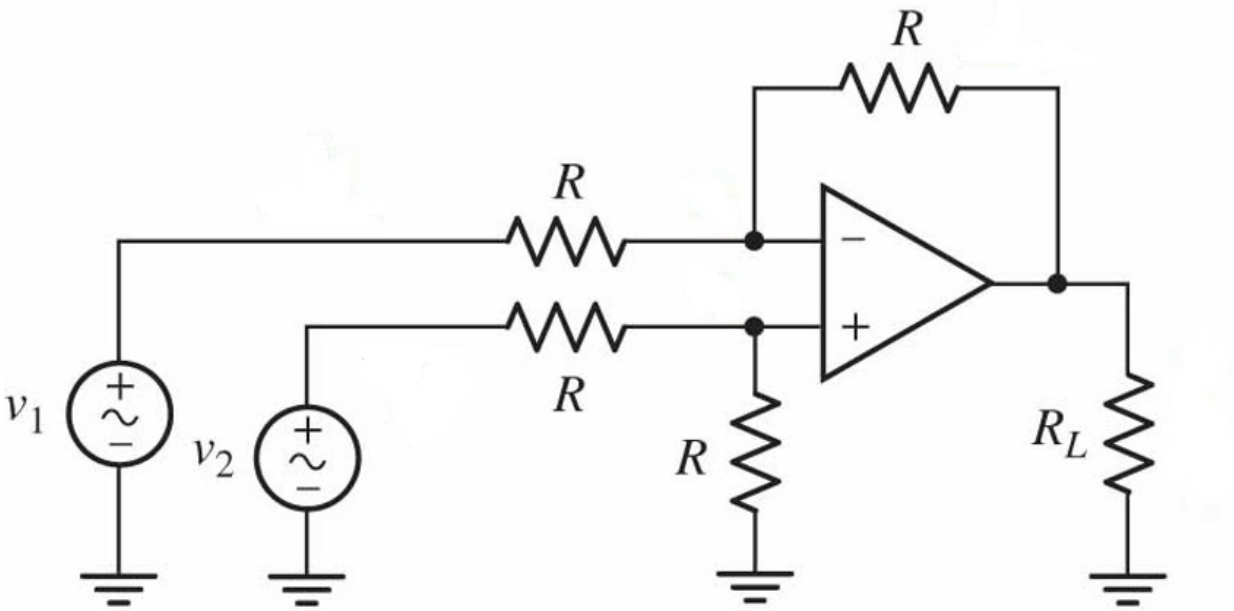
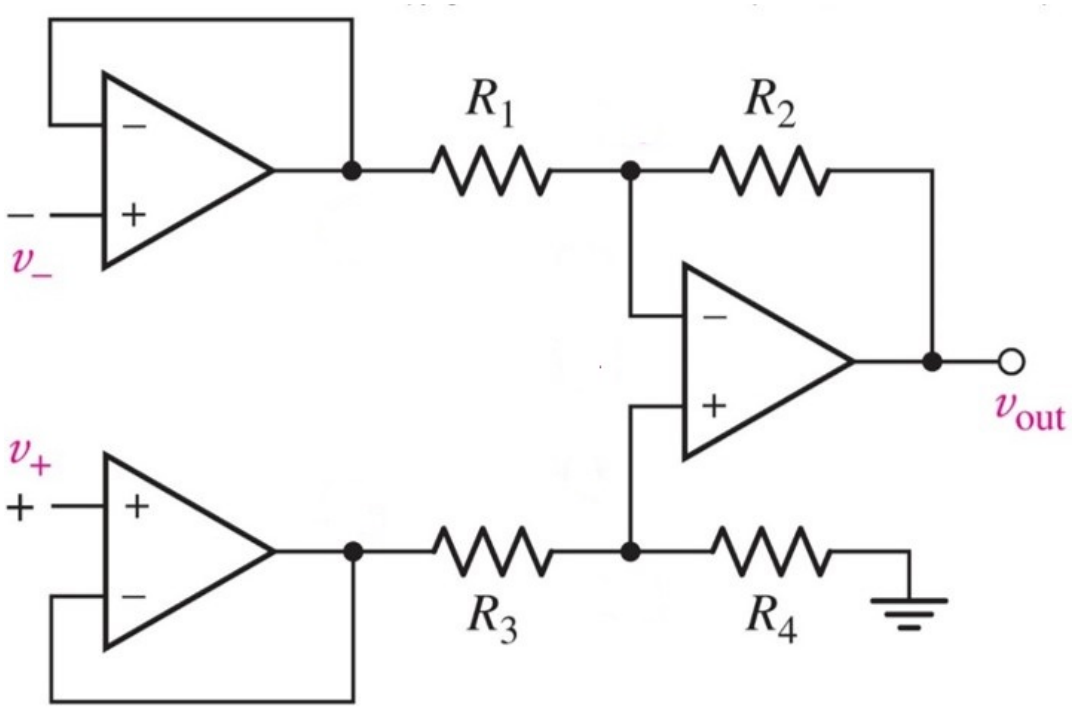
Tutorial question 1

Derive an expression for v_{out} in terms of v_1 and v_2 for the circuit shown below, which is known as a difference amplifier.



Tutorial question 2

If $R_4/R_3 = R_2/R_1 = K$, derive an expression for v_{out} in terms of v_+ and v_- for the circuit shown below, which is known as the instrumentation amplifier.



Tutorial question 3

Design a circuit using two ideal op amps to produce the following output relationship:

$$v_{out} = 2v_1 + 3v_2$$

Assume both op amps operate in the linear region. Use an inverting summing amplifier followed by an inverting amplifier. All resistors must lie between 10 kΩ and 100 kΩ.

(a) Draw your circuit and label all resistor values and the intermediate output v_x.

(b) Derive expressions for both stage outputs to verify your design.

(c) Calculate both stage outputs when $v_1 = 0.5$ V and $v_2 = -0.2$ V.

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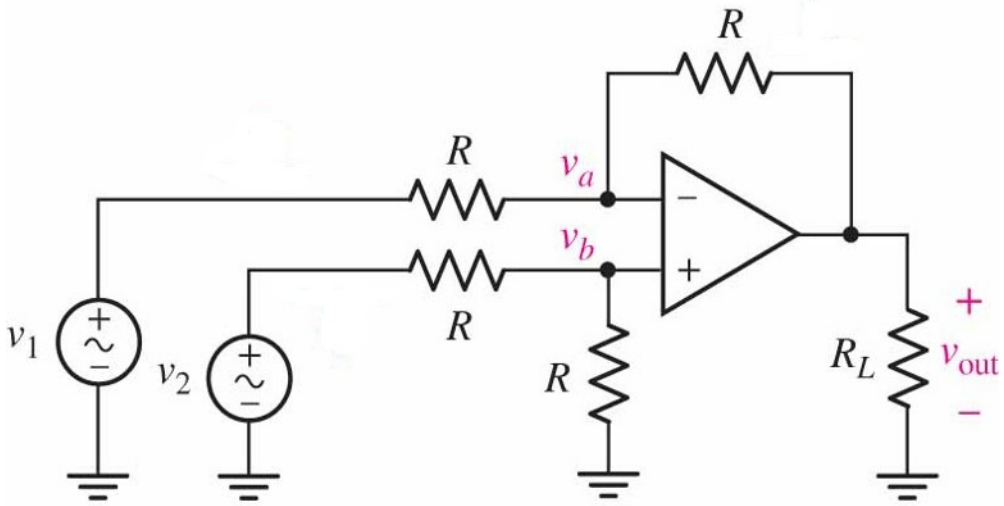
Solution to Tutorial question 1

Apply KCL to node v_a,

$$\frac{v_a - v_1}{R} + \frac{v_a - v_{out}}{R} = 0 \qquad [1]$$

Apply KCL to node v_b,

$$\frac{v_b - v_2}{R} + \frac{v_b}{R} = 0 \qquad [2]$$



Combing Eq. [1] and [2], and according to op amp rule 2 ($v_a = v_b$), we obtain

$$v_{out} = v_2 - v_1 \qquad [3]$$

Solution to Tutorial Question 2

- According to op amp rule 1, at the node labeled v_x:

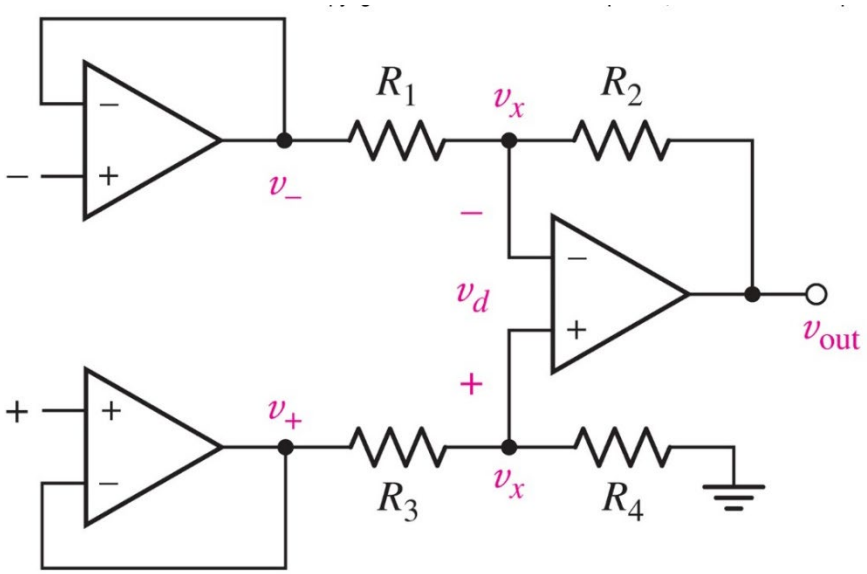
$$\frac{v_x - v_-}{R_1} + \frac{v_x - v_{out}}{R_2} = 0 \qquad [1]$$

- and

$$\frac{v_x - v_+}{R_3} + \frac{v_x}{R_4} = 0 \qquad [2]$$

- Solving Eq.[2], we find v_x:

$$v_x = \frac{v_+}{1 + R_3/R_4} \qquad [3]$$



- Substitute Eq.[3] into Eq.[1], we obtain v_{out}:

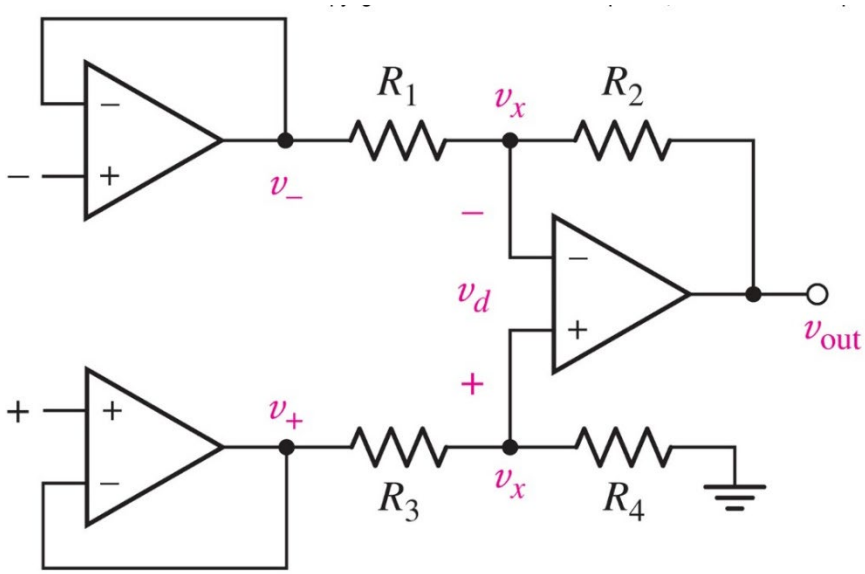
$$v_{out} = \frac{R_4}{R_3}\left(\frac{1 + R_2/R_1}{1 + R_4/R_3}\right) v_+ - \frac{R_2}{R_1} v_- \qquad [4]$$

Solution to Tutorial Question 2

- Since $R_4/R_3 = R_2/R_1 = K$, the Eq.[4] reduces to

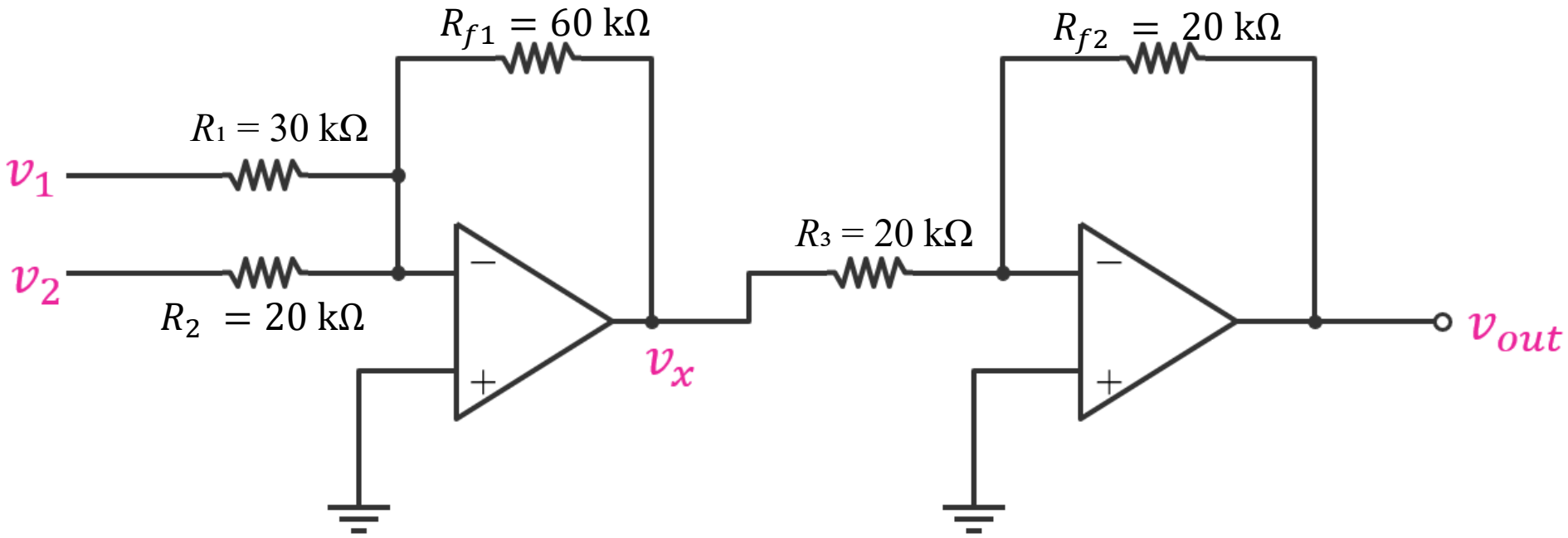
$$v_{out} = K(v_+ - v_-) \qquad [5]$$

- In this case, only difference is amplified, and the gain K is set by resistors ratio.



Solution to Tutorial question 3

(a) One valid design is shown below.



$R_1 = 30\ \text{k}\Omega,$ $R_2 = 20\ \text{k}\Omega,$ $R_{f1} = 60\ \text{k}\Omega$

$R_3 = 20\ \text{k}\Omega,$ $R_{f2} = 20\ \text{k}\Omega$

The non-inverting input of each op amp is grounded.

Solution to Tutorial question 3

(b) The first stage is an inverting summing amplifier:

$$v_x = -\frac{R_{f1}}{R_1}v_1 - \frac{R_{f1}}{R_2}v_2 = -2v_1 - 3v_2$$

The second stage is an inverting amplifier:

$$v_{out} = -\frac{R_{f2}}{R_3}v_x = -\frac{20\ \text{k}\Omega}{20\ \text{k}\Omega}v_x = -v_x$$

$$v_{out} = 2v_1 + 3v_2$$

(c) For the specified input voltages:

$$v_x = -2(0.5) - 3(-0.2) = -0.4\ V$$

$$v_{out} = -v_x = 0.4\ V$$

Other resistor values are acceptable if they satisfy the design requirements.